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Anderson et al.

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(54) **DEVICE FOR INCREASING VISIBILITY OF GUY LINES AT NIGHT**

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E04H 15/10 (2006.01)

F21L 4/08 (2006.01)

F21V 23/04 (2006.01)

F21Y 115/10 (2016.01)

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CPC **F21V 21/0885** (2013.01); **E04H 15/10** (2013.01); **F21L 4/08** (2013.01); **F21V 23/0414** (2013.01); **F21Y 2115/10** (2016.08)

(58) **Field of Classification Search**

CPC **F21V 21/0885**; **F21V 23/0414**; **F21L 4/08**; **F21Y 2115/10**

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,740,982 A * 12/1929 Hillier F16G 11/105
24/115 M

5,707,135 A 1/1998 Miller, Jr.
(Continued)

FOREIGN PATENT DOCUMENTS

WO 2015120332 A1 8/2015

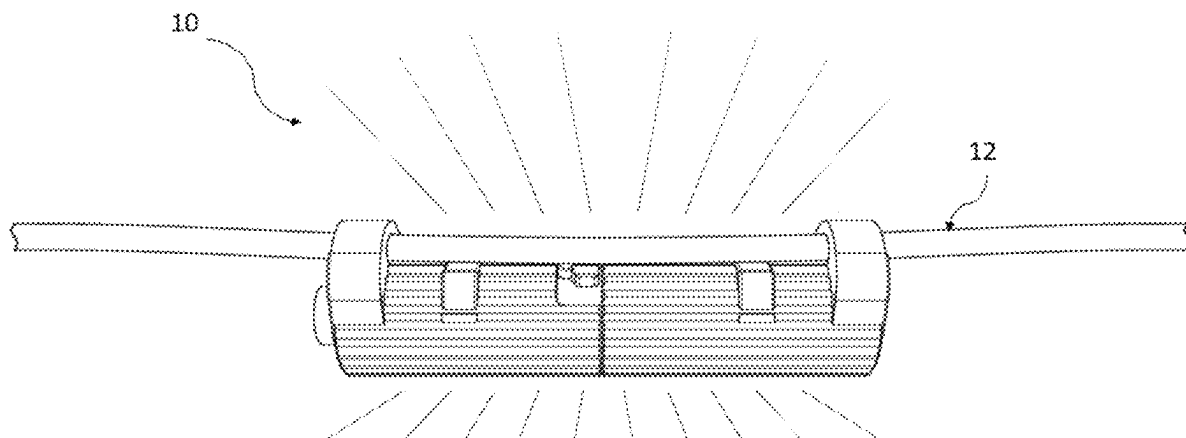
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(57) **ABSTRACT**

Currently disclosed is a device for increasing visibility of a guy line at night. The device includes a housing, a light emitting source, a power source, and an on/off switch. The housing having an internal portion, external portion, and an attachment mechanism located on the external portion of the housing locking in place the housing to the guy line. The light emitting source is located within the housing and powered by the power source. The on/off switch is in contact with the power source providing readily activation of the light emitting source. The attachment mechanism may comprise at least one hook portion and a tensioning mechanism. The tensioning mechanism may comprise at least one offset portion. The tensioning mechanism prevents the device from sliding down the guy line.

17 Claims, 9 Drawing Sheets



Related U.S. Application Data

continuation of application No. 17/587,557, filed on
Jan. 28, 2022, now Pat. No. 11,530,801.

- (60) Provisional application No. 63/143,272, filed on Jan.
29, 2021.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,353,946	B1	3/2002	Steiner	
6,758,588	B2 *	7/2004	Hsu	B60Q 1/56 362/582
7,021,809	B2	4/2006	Iwasa	
7,410,271	B1	8/2008	Man	
7,544,082	B1 *	6/2009	Halvorsen	H01R 13/5213 439/367
2003/0223247	A1	12/2003	Karras	
2007/0279902	A1	12/2007	Kim	
2009/0173455	A1	7/2009	Hartley	
2012/0140457	A1	6/2012	McQuade	
2013/0298845	A1	11/2013	Blanchard	
2016/0033117	A1	2/2016	Grigore	
2016/0168815	A1	6/2016	Kemp	
2017/0138585	A1	5/2017	Korenek	
2021/0140625	A1	5/2021	Holliger	

* cited by examiner

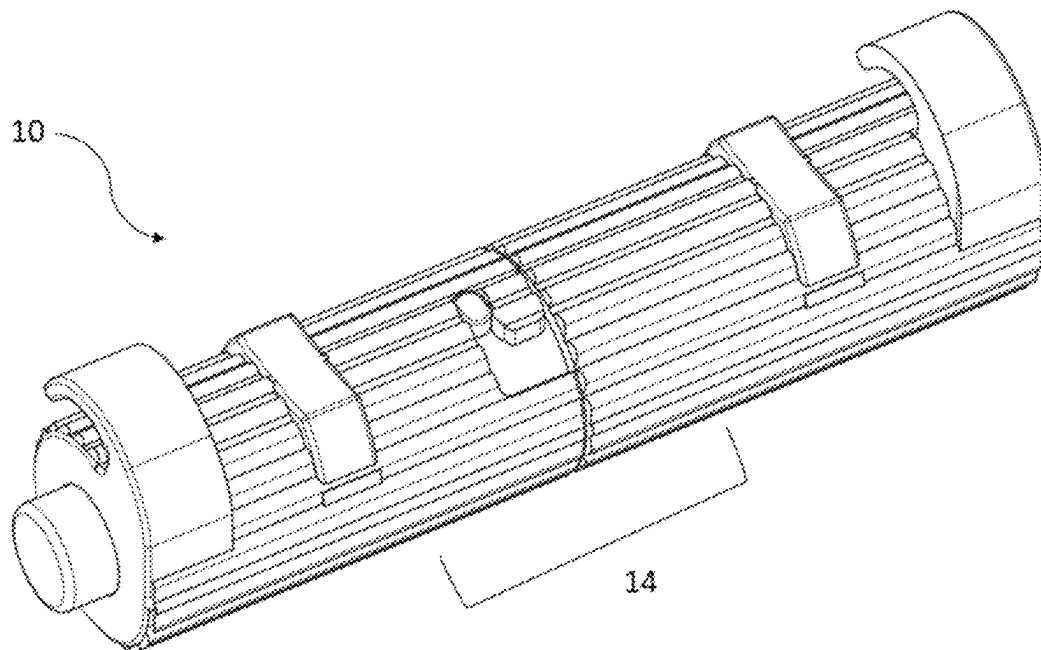


Fig. 1

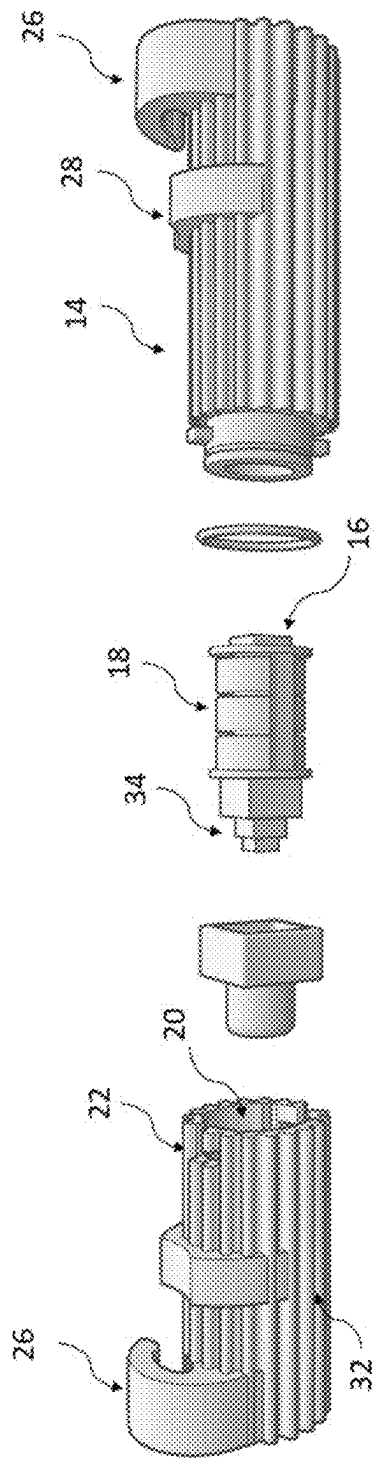


Fig. 2

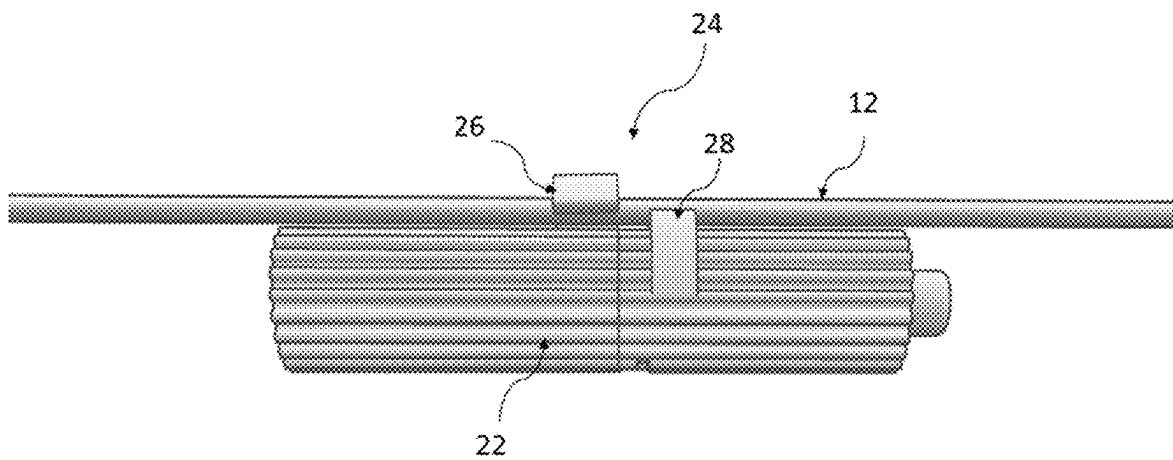


Fig. 3

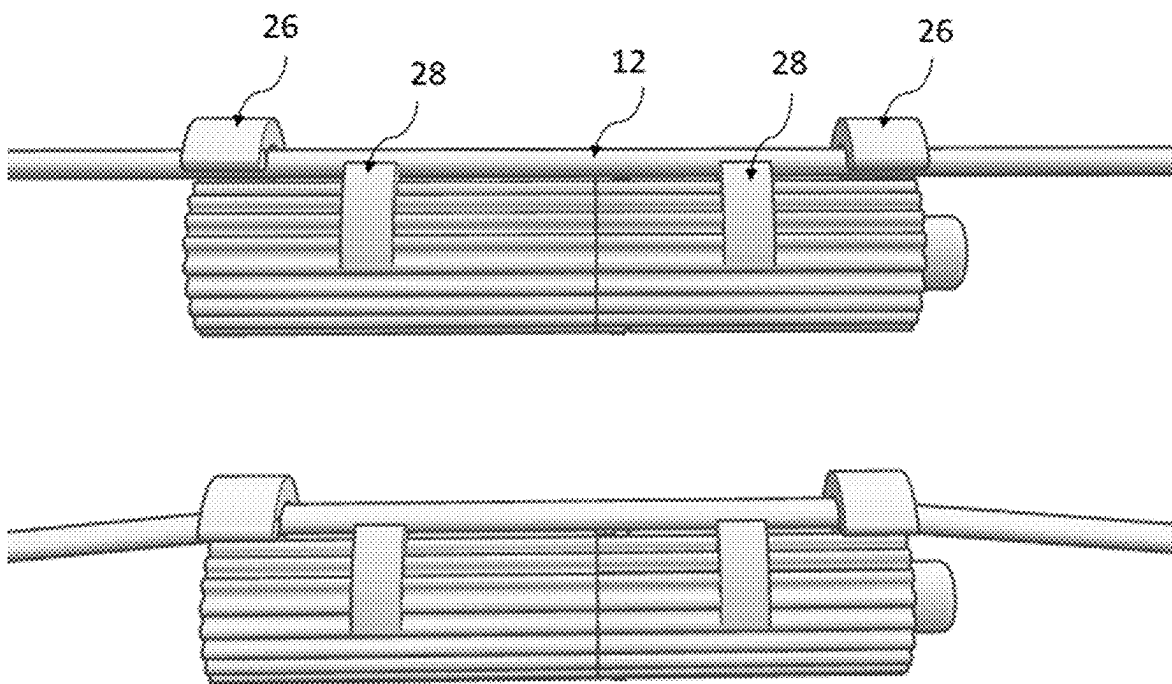


Fig. 4

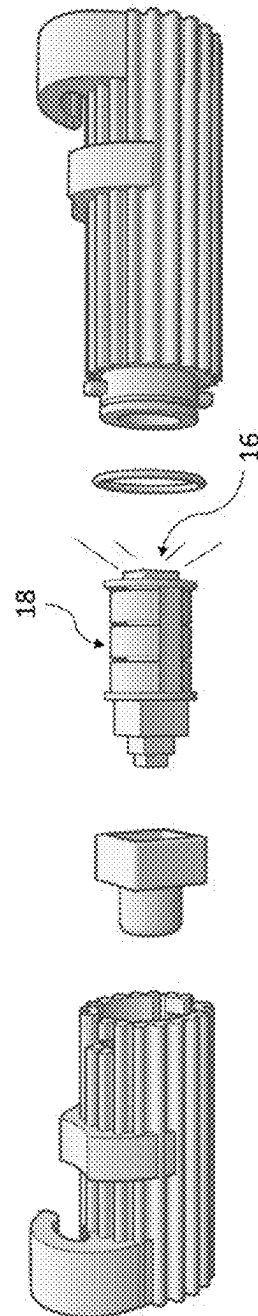


Fig. 5

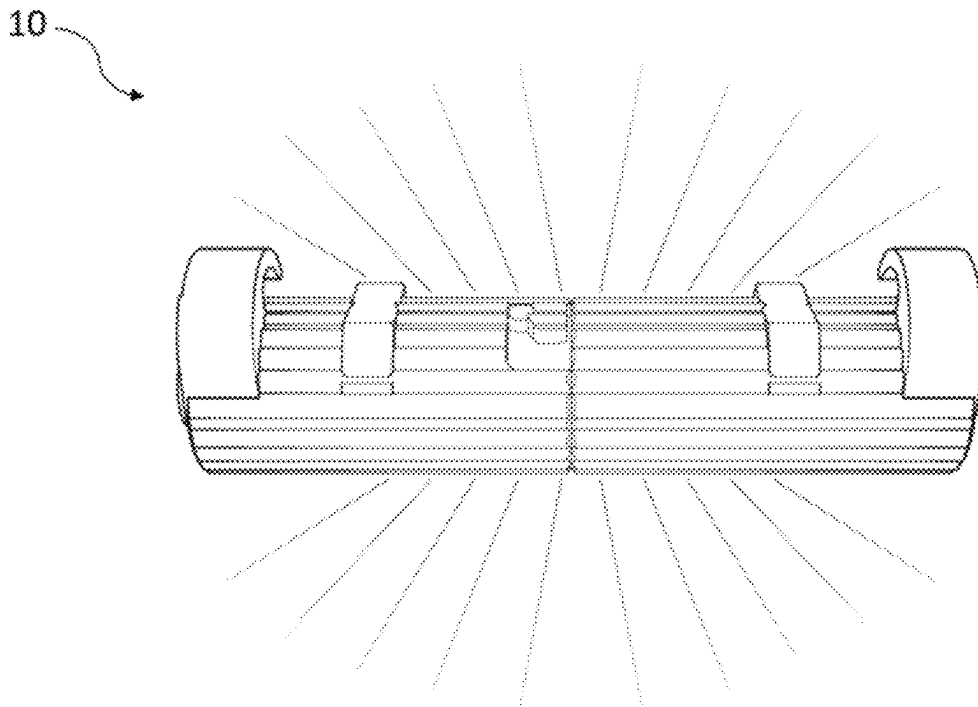


Fig. 6

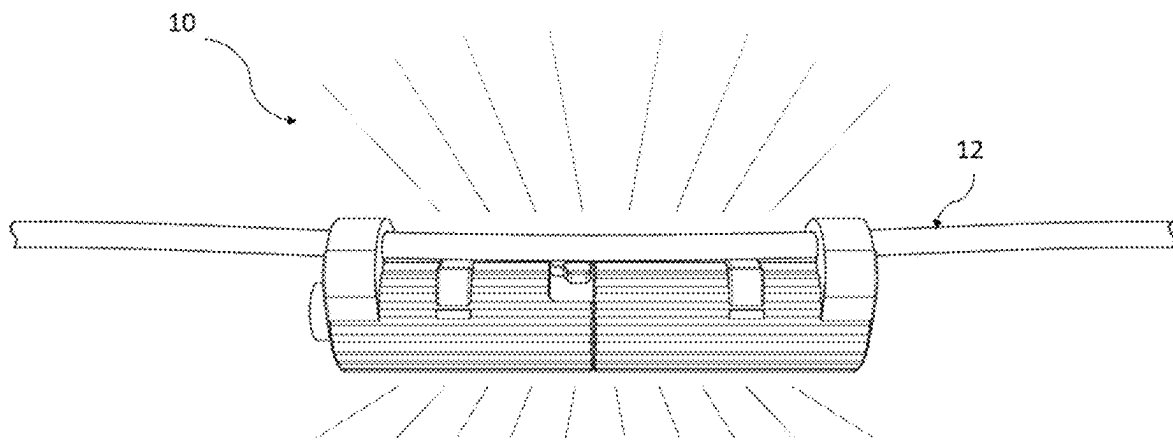


Fig. 7

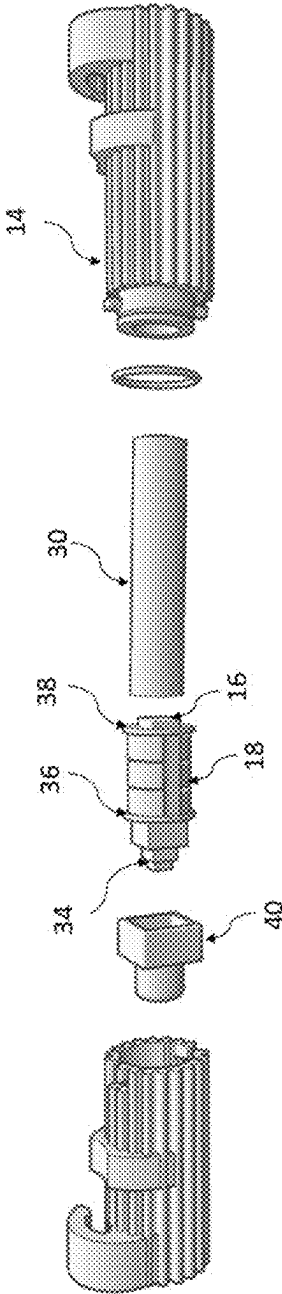


Fig. 8

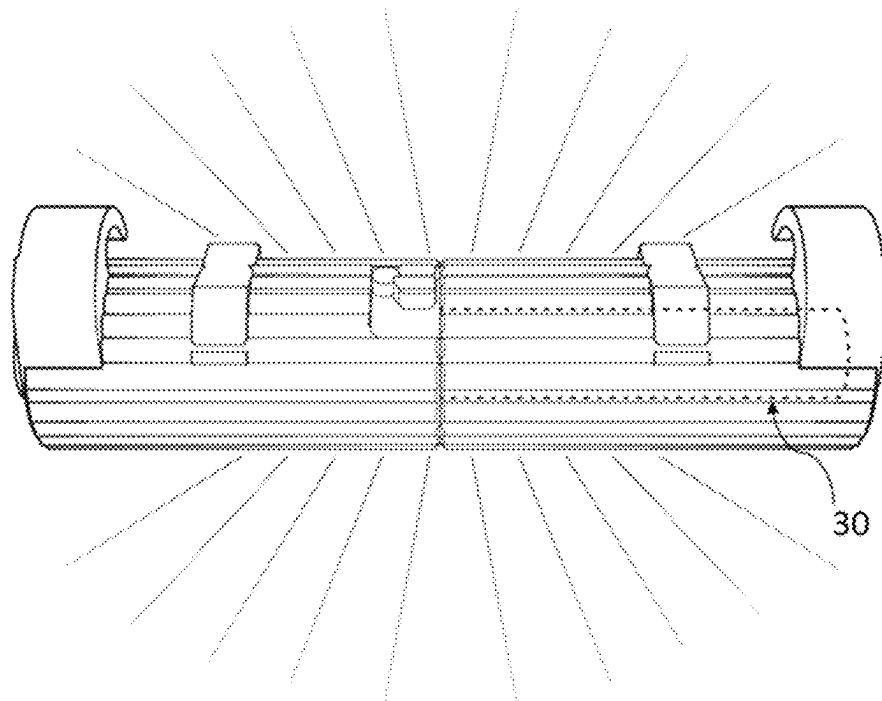


Fig. 9

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DEVICE FOR INCREASING VISIBILITY OF GUY LINES AT NIGHT

This application is a continuation of and claims priority to U.S. patent application Ser. No. 17/988,288 entitled “Device for Increasing Visibility of Guy Lines at Night” filed Nov. 16, 2022, which is a continuation of and claims priority to U.S. patent application Ser. No. 17/587,557 entitled “Device for Increasing Visibility of Guy Lines at Night” filed Jan. 28, 2022, now issued as U.S. Pat. No. 11,530,801, which claims benefit of U.S. Provisional Patent Application Ser. No. 63/143,272, filed on Jan. 29, 2021, each of which are incorporated by reference in their entirety herein.

BACKGROUND

Most modern tents used in recreational camping are supported with ropes or wires attached to stakes driven into the ground. Said ropes or wires are commonly referred to as guy wires or guy lines. Guy lines are typically taunt angled low to the ground providing support and stability to a freestanding structure, such as a tent. However, guy lines may cause problems when navigating around the tent or other structure as people can trip on the taunt guy lines. This is especially a problem at night, as the stakes and lines around a tent are farther out than the perimeter of the tent itself. Additionally, due to their small size, guy lines are near invisible at night making them hazardous. Therefore, there is a need for a solution for increasing the visibility of guy lines at night to reduce the probability of someone tripping over the lines and potentially becoming injured.

SUMMARY

Currently disclosed is a safety device for increasing visibility of a guy line at night. The device comprises a housing, a light emitting source, a power source, and an on/off switch. The housing comprising an internal portion, external portion, and an attachment mechanism located on the external portion of the housing locking in place the housing to the guy line. The light emitting source is located within the housing and powered by the power source. The on/off switch is in contact with the power source and provides for readily activation of the light emitting source.

The attachment mechanism may comprise of at least one hook portion and a tension mechanism, wherein the tension mechanism comprises at least one offset portion. The at least one hook portion attaches to the guy line. The offset portion pushes the guy line into the hook portion providing tension to the guy line and locking in place the device to the guy line; thus, preventing the device from sliding down the guy line. The device may be held in place at any point along the length of the guy line.

BRIEF DESCRIPTION OF THE DRAWINGS

Further features of the inventive embodiments will become apparent to those skilled in the art to which the embodiments relate from reading the specification and claims with reference to the accompanying drawings, in which:

FIG. 1 illustrates a device for increasing visibility of a guy line;

FIG. 2 illustrates an exploded view of a device for increasing visibility of a guy line;

FIG. 3 illustrates a device for increasing visibility of a guy line;

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FIG. 4 illustrates a device for increasing visibility of a guy line;

FIG. 5 illustrates a light emitting source in an exploded view of a device for increasing visibility of a guy line;

FIG. 6 illustrates an illuminated device for increasing visibility of a guy line;

FIG. 7 illustrates an illuminated device for increasing visibility of a guy line;

FIG. 8 illustrates an exploded view of a device for increasing visibility of a guy line; and

FIG. 9 illustrates an illuminated device for increasing visibility of a guy line.

DETAILED DESCRIPTION OF THE DRAWINGS

As illustrated in FIGS. 1-9, currently disclosed is a device 10 for increasing visibility of a guy line 12 at night. The device comprises a housing 14, a light emitting source 16, a power source 18, and an on/off switch 34. The housing 14 comprises an internal portion 20, an external portion 22, and an attachment mechanism 24. The attachment mechanism 24 is located on the external portion 22 of the housing. The light emitting source is located within the housing and is powered by the power source. The on/off switch 34 is in contact with the power source 18.

The attachment mechanism locks in place the housing to the guy line. In one embodiment, as shown in FIG. 3, the attachment mechanism may comprise at least one hook portion 26 and a tension mechanism, wherein the tension mechanism comprises at least one offset portion 28. As illustrated in FIG. 4, the attachment mechanism may comprise two hook portions 26 and a tension mechanism, wherein the tension mechanism comprises two offset portions 28, each offset portion located in close proximity to a hook portion 26. In another embodiment, the attachment mechanism may comprise a screw down mechanism. In yet another embodiment, the attachment mechanism may comprise a clamping mechanism, for example, a clip/flip mechanism or clothespin mechanism. In another embodiment, the attachment mechanism may comprise a chemical attachment mechanism, for example, adhesive.

As indicated above, the attachment mechanism may comprise at least one hook portion 26 and a tension mechanism, wherein the tension mechanism comprises at least one offset portion 28. As illustrated in FIG. 4, the hook portion 26 attaches to the guy line 12. The offset portion 28 pushes the guy line 12 into the hook portion 26 providing tension to the guy line locking in place the device to the guy line; thus, preventing the device from sliding down the guy line. In this manner, the device 10 can be held in place at any point along the length of the guy line 12. For example, the device may be located and held in place at the bottom of the guy line near the stake driven into the ground, or the device may be located and held in place at the center of the guy line. Alternatively, multiple devices 10 may be placed along the length of a single guy line to further increase the visibility of the guy line at night.

As shown in the Figures, the device comprises a light emitting source 16, for example a light emitting diode (LED). As illustrated in FIG. 5, the light emitting source 16 is located within the housing 14 and powered by power source 18. As illustrated in FIGS. 5-7, the light emitting source shines light throughout the length of the housing creating an illuminating effect. When the device is placed on a guy line, said illuminating effect draws attention to the presence of the guy line.

In a preferred embodiment, the light emitting source dimly illuminates the housing; therefore, extending the power source life of the device. Furthermore, the dimmed light provides a comfortable amount of light to users. For example, the illuminated device would not pose a bothersome amount of light that would keep campers awake at night or reduce nighttime visibility by blinding the user. Moreover, the dimmed light prevents insects from being attracted to the illuminated device.

As shown in the Figures, housing **14** comprises internal portion **20**, external portion **22**, and attachment mechanism **24** located on the external portion. The housing may further comprise two removably attached portions, wherein the light emitting source and power source are located on one of the two removably attached portions. Further, the exterior portion of the housing may comprise a ribbed surface **32**. The ribbed surface provides for light from the light emitting source to be diffused from the interior portion of the housing to the exterior portion of the housing preventing sharp light points from the illuminated device.

In another embodiment, the device **10** may further comprise a tubular body within the housing. As illustrated in FIGS. **8-9**, tubular body **30** is located within the housing **14** following the light emitting source **16**. The tubular body further extends the light from the light emitting source throughout the housing **14**. The tubular body may be removably attached to the interior portion of the housing. Further, the tubular body may comprise of malleable plastic material.

As discussed above, housing **14** may further comprise two removably attached portions, wherein the light emitting source and power source are located on one of the two removably attached portions. As illustrated in FIG. **9**, in an embodiment wherein the device **10** further comprises tubular body **30**, the light emitting source and power source are located on one of the two removably attached portions (not shown) and the tubular body is located on the second of the two removably attached portions (shown in dashed lines).

As claimed, the device **10** comprises in part of power source **18**. The power source **18** is located within the housing **14** and powers the light emitting source **16**. The power source may comprise a battery, for example, a cell battery. Alternatively, or additionally, the power source may comprise a solar cell battery. In another embodiment, the device **10** may further comprise a light sensor to detect the ambient relative light level. When the light level is below a certain level, the device may automatically turn the light emitting source on. Correspondingly, when the light level is above a certain level, the device may automatically turn the light emitting source off.

The device **10** comprises in part an on/off switch **34**. The on/off switch **34** is in contact with the power source **18** providing for the light emitting source to be turn on/off, as desired. In an embodiment, as illustrated in FIG. **8**, the on/off switch **34** may be connected to a first electrically conductive PCB **36** located within the housing, said electrically conductive PCB **36** in electrical contact with the power source **18**, and a second electrically conductive PCB **38** in electrical contact with the light emitting source **16**. When the on/off switch **34** is activated, the circuit is completed. The light emitting source is, in turn, activated and the device **10** is illuminated. The on/off switch may be a self-locking switch. Further, the on/off switch may be covered by a switch cover **40**, wherein said cover may be a silicone switch cover.

While this invention has been shown and described with respect to a detailed embodiment thereof, it will be understood by those skilled in the art that changes in form and

detail thereof may be made without departing from the scope of the claims of the invention.

What is claimed:

1. A device for increasing visibility of a guy line at night comprising:

a housing comprising an internal portion, an external portion, and an attachment mechanism;

a light emitting source;

a tubular body;

a power source; and

an on/off switch,

wherein the attachment mechanism is located on the external portion of the housing for locking in place the housing to a guy line positioned along a length of the housing,

wherein the light emitting source is located within the housing,

wherein the tubular body is configured in the internal portion of the housing,

wherein the tubular body diffuses and extends light generated by the light emitting source throughout the housing,

wherein the light emitting source is powered by the power source, and wherein the on/off switch is in contact with the power source.

2. The device as claimed in claim 1, wherein the on/off switch is connected to a first electrically conductive PCB located within the housing, said electrically conductive PCB in electrical contact with the power source, and a second electrically conductive PCB, said second electrically conductive PCB in electrical contact with the light emitting source.

3. The device as claimed in claim 1, wherein the attachment mechanism comprises at least one hook portion and a tensioning mechanism, wherein the tensioning mechanism comprises at least one offset portion.

4. The device as claimed in claim 1, wherein the attachment mechanism comprises two hook portions and a tensioning mechanism, wherein the tensioning mechanism comprises two offset portions.

5. The device as claimed in claim 1, wherein the attachment mechanism comprises a screw down mechanism.

6. The device as claimed in claim 1, wherein the attachment mechanism comprises a clamping mechanism.

7. The device as claimed in claim 1, wherein the attachment mechanism comprises an adhesive.

8. The device as claimed in claim 1, wherein the light emitting source comprises a light emitting diode.

9. The device as claimed in claim 1, wherein the housing further comprises a two part elongated body having a first end and a second end opposite the first, wherein the two part elongated body comprises two removably attached portions.

10. The device as claimed in claim 1, wherein the external portion of the housing is a ribbed surface facilitating light diffusion.

11. The device as claimed in claim 1, wherein the power source comprises a cell battery.

12. The device as claimed in claim 1, wherein the power source further comprises a solar cell battery.

13. The device as claimed in claim 1, further comprising a light sensor in contact with the power source.

14. The device as claimed in claim 1, wherein the housing is made out of a translucent plastic material.

15. The device as claimed in claim 1, wherein the on/off switch is a self-locking switch.

16. The device as claimed in claim 15, wherein the self-locking switch is located within the internal portion of

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the housing and positioned closer to the first end than the second end of the housing and a switch cover is a silicone switch cover.

17. The device as claimed in claim 1, wherein the on/off switch is actuatable through a switch cover protruding from the internal portion of the housing and located at the first end of the housing.

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